

Similarly, wireless connectivity has moved from peripheral to standard. This includes support for Wi-Fi 6, LTE, and 5G across devices, with technologies such as M.2, including Jetson Orin, UDOO Bolt, and ASRock Industrial offerings.

In latency-constrained robotic systems, particularly those built on ROS and ROS 2, the interaction between I/O topology, camera pipelines, networking, and software frameworks is as critical as raw compute capability. As demonstrated by platforms targeting autonomous mobile robots and inspection systems, scalability increasingly depends on how efficiently data enters and moves through the SBC, not just how quickly it is processed.

Why India is a natural fit for the new SBC era

Smart manufacturing, infrastructure modernisation, and the use of AI-driven technologies align closely with present-day SBC adoption trends. Compact industrial SBCs are being used for vision-based quality checks, predictive maintenance, and real-time machine monitoring on existing manufacturing lines, incorporating smart methodologies into legacy infrastructure.

In railways and transportation, rugged fanless SBCs are replacing traditional PCs in monitoring and passenger information systems. The rollout of EV charging infrastructure and kiosks is also driving demand for secure, always-on platforms featuring secure boot, Trusted Platform Modules, and remote management, making Atom SBCs a strong fit.

Another rapidly growing field is robotics, with Indian robotics start-ups developing AMRs, inspection robots, and collaborative robotic systems using SBC technologies from ADLINK, Nvidia Jetson partners, and AAEON Technology Corporation in ROS-based environments.

In medical electronics, service life cycle and OS stability matter far

more than maximum performance capability, making industrial-grade SBCs from ADLINK or Advantech strong candidates.

The SBC is no longer a component—It's a platform decision

A decade ago, choosing an SBC meant checking processor speed, board size, and price. Today, that approach almost guarantees painful redesigns. As Connie Hsiao, Asia Pacific Marketing Manager, and Dave Yu, Business Development Manager/AI Edge Computing, at ADLINK Technology explain, “the company’s focus is on building scalable edge AI platforms aligned with the Intel roadmap, spanning Atom, Core Ultra, and next-generation architectures, so customers can future-proof their designs. Customers no longer want fixed platforms. They want configurable systems, low power or high performance, minimal I/O or sensor-heavy, so our SBCs are designed to be tailored, not one-size-fits-all.”

Integrated AI acceleration, thermal resilience, modular expansion, software longevity, security features such as TPM and secure boot, and life cycle alignment now determine whether a system survives in the real world.

Future SBC designs will integrate higher AI compute, expanded MIPI camera support, and readiness for next-generation AI models, enabling faster deployment of vision-based edge applications. More than anything else, this reflects what is truly happening within the SBC world: the SBC has evolved from a simple controller into the foundation for intelligent, resilient, and scalable edge solutions. **EFY**

This article is based on industry inputs from Venkatesh K., Application Engineer, Dave Yu, Business Development Manager/AI Edge Computing, and Connie Hsiao, Asia Pacific Marketing Manager, at ADLINK, complemented by independent research and analysis. It has been curated by Akanksha Sondhi Gaur, Senior Technology Journalist at EFY, combining firsthand insights with secondary research.

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-Jeff Bezos

